Materializing Morality

Design Ethics and Technological Mediation

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abstract:
During the past decade, the 'script' concept, indicating the actively prescribing role of technologies in human actions, has acquired a central place in STS. Up to now, the concept has mainly functioned in *descriptive* settings. This paper will deploy the concept in a *normative* setting. When technologies play a role in the coming about of human actions, they give material answers to the ethical question 'how to act?'. This implies that engineers are doing 'ethics with other means:' they materialize morality. The paper will explore the implications of this insight for engineering ethics. It first augments the concept, by developing the notion of 'technological mediation' and showing that scripts can be seen as a form of mediation. After this, it investigates how the concept of mediation could be made fruitful for design ethics. It does so by discussing how the ambition to design technologies with the explicit aim to influence human actions raises moral questions itself, and by elaborating two methods for anticipating technological mediation in the design process: 'anticipation by imagination' in the form of a 'mediation analysis'; and an augmented version of constructive technology assessment.

keywords:
philosophy of technology, engineering ethics, technology assessment, design

1. Conceptualizing the moral dimension of technologies

The ethics of engineering design aims to analyze the moral aspects of technology design in a systematic way. One of the things that should be taken into account in such analyses is the social impact that the technology-in-design will have as soon as it enters society. As recent research in Science and Technology Studies and the Philosophy of Technology has shown, technologies profoundly influence the behavior and experiences of users. This charges the ethics of engineering design with the task to conceptualize this influence, and to anticipate it in design.

A common sense vision of the role of technologies in society is guided by the concept of functionality. Technologies are

designed to function. They should solve a problem or fill a need. In this view, the criteria by which technologies are to be assessed primarily concern their ability to realize the ends to which they were designed. From this perspective, the focus of engineering ethics can only concern the normative aspects of the *goals* for which technologies are designed, or the quality of the way in which the technologies *function* (risk assessment).

The script concept, as developed about a decade ago by Madeleine Akrich (1992) and Bruno Latour (1992), challenges this strictly functional vision of technology. Akrich and Latour introduced this concept in order to describe the manifold roles technological artifacts play in their use contexts. Like a theater play or a movie, they hold, technologies possess a 'script' in the sense that they can prescribe the actions of the actors involved. Technologies are able to evoke certain kinds of behavior: a speed bump can invite drivers to drive slowly, because of its ability to damage the shock absorbers; a car can demand from a driver that he or she wear the safety belt by refusing to start if the belt is not used; a plastic coffee cup has the script 'throw me away after use,' whereas a porcelain cup 'asks' to be cleaned and used again. Technological artifacts can influence human behavior, and this influence can be understood in terms of scripts.

Scripts, according to Akrich and Latour, are the products of 'inscriptions' by designers. Designers anticipate how users will interact with the product they are designing and, implicitly or explicitly, build prescriptions for use into the materiality of the product. Latour describes this inscription process in terms of 'delegation:' designers delegate specific responsibilities to artifacts, like the responsibility to make sure nobody drives too fast, which is delegated to a speed bump.

Conceptualizing technological artifacts in terms of scripts shows that functionality is too limited a concept for engineering ethics. Scripts transcend functionality: they form a surplus to

it, which occurs once the technology is functioning. When technologies fulfill their functions, they also help to shape the actions of their users. For this reason, the script concept lays bare ethical questions regarding technology design that transcend the common sense idea that technologies only need to be morally evaluated in terms of the goals for which they are designed or of the quality of their functioning. The script approach opens up a new way to morally assess technologies with respect to the role they play in their use contexts. In doing so, it also reveals a specific responsibility of the designer, who can be seen as the 'inscriber' of scripts.

Up to now, the script concept has mainly functioned in *descriptive* settings: in analyses of the role of technologies in their use contexts. In this article, I investigate how the concept can be deployed in a *normative* setting. In order to cover all relevant aspects of the role of technological artifacts in their use contexts and to provide a vocabulary for describing these aspects, I will first develop the notion of 'technological mediation' and show that scripts can be seen as a specific aspect of mediation. After this, I will investigate how the concept of mediation can be made fruitful for design ethics. The concept provides designers with an explicit link with the use context in which their products will be functioning, making it possible to identify and answer new normative questions during the design process.

Integrating mediation in engineering ethics is a complex task, however. Firstly, the ambition to design technologies with the explicit aim to influence human actions raises moral questions itself. It is not self-evident, after all, that all attempts to steer human behavior are morally justified, and steering human beings with the help of technology raises associations with the totalitarian technocracy of Orwell's Big Brother. Secondly, if some forms of behavior-steering technologies can be seen as morally acceptable -- and I think such technologies do exist --

it is very complicated to design them, since there is no linear connection between the activities of designers and the mediating role of the artifacts they are designing. As I will make clear, this mediating role also depends on the unpredictable ways in which the technologies are used. For this reason, I will suggest two ways to cope with this unpredictability.

2. Technological mediation[1]

For analyzing the role of technologies in the daily lives of human beings, phenomenology offers a suitable framework -- at least when the romantic resistance of classical phenomenology against science and technology is left behind and when phenomenology is defined broadly as *the philosophical analysis of human-world relationships* (Cf. Verbeek 2000, pp. 119-136). The central idea in the phenomenological approach that I will take as a starting point in this article is that subject and object -- or humans and reality -- mutually constitute each other. Humans and reality are always interrelated, phenomenology holds. Human beings cannot but be directed at the world around them; they are always experiencing it and it is the only place where they can realize their existence. Conversely, their world can only be what it is when humans deal with it and interpret it. In their interrelation, both the subjectivity of humans and the objectivity of their world take shape. What human beings are and what their world is, is co-determined by the relations and interactions they have with each other.

This phenomenological perspective offers a framework for analyzing the influence of technology on human behavior, since the role of technology in people's every day lives can be located in the very relation between humans and world that is central in phenomenology. Technological artifacts mediate how human beings are present in their world, and how their world is present to them.

human-technology relationships

A good starting point for understanding technological mediation is Don Ihde's analysis of the relationships between humans and technological artifacts (Don Ihde, 1990). Ihde discerns several relationships human beings can have with technologies. Firstly, Ihde discerns the 'embodiment relation,' in which technologies are 'incorporated' by their users, making possible a relationship between humans and their world 'through' the technological artifact. This embodiment relation, for instance, occurs when looking through a pair of glasses; the artifact is not perceived itself, but helps to perceive the environment. Technological artifacts become extensions of the human body here, as it were. Secondly, Ihde discerns the 'alterity relation,' in which technologies are the *terminus* of our experience. This relation occurs when interacting with a device as if it were another living being, for instance when buying a train ticket at an automatic ticket dispenser. Thirdly, Ihde discerns the 'background relation.' In this relation, technologies play a role at the background of our experience, creating a context for it.[2] An example of this relation is the automatic switching on and off of the refrigerator. The figure below illustrates these three human-technology relations:

embodiment relation	(human – technology) → world
alterity relation	human → technology (– world)
background relation	human (– technology – world)

Figure 1: Human-technology relationships (after Ihde, 1990)

For understanding the influence technological artifacts can have on human behavior, the embodiment relation is the most important.

Ihde's analysis of this relation is based on Martin Heidegger's analysis of the role of tools in the everyday relation between humans and their world. According to Heidegger (1927), tools should not simply be understood as functional instruments, but as 'connections' or 'linkages' between humans and reality. The way in which tools are present to human beings when they are used, Heidegger calls 'readiness-to-hand.' Tools that are used for doing something, typically withdraw from people's attention. A person who drives a nail into a wall, is not directed at the hammer, but at the nail. People's involvement with reality takes place *through* the ready-to-hand artifact. Only when it breaks down, it asks attention for itself again. The artifact is then, in Heidegger's words, 'present-at-hand' and is not able to facilitate a relationship between a user and his or her world anymore. This 'presence-at-hand' can be seen as the basis for Ihde's 'alterity relation,' just as the 'embodiment relation' is based on Heidegger's 'readiness-to-hand.'

Heidegger's analysis of readiness-to-hand is of utmost importance for a 'phenomenology of things'. Ready-to-hand artifacts withdraw from people's attention but nevertheless they do play a constitutive role in the human-world relation that arises around them. When using a technological artifact, it facilitates people's involvement with reality, and in doing so it co-shapes how humans can be present in their world and their world for them. Things-in-use should be understood as *mediators* of human-world relationships. Mediation should be understood in an active sense here. Artifacts are not neutral intermediaries, but actively co-shape people's being in the world: their perceptions and actions, experience and existence.

The positions of Don Ihde and the French philosopher and anthropologist Bruno Latour offer concepts for building a vocabulary to gain a closer understanding this mediating role of technologies. In order to build this vocabulary, I discern two directions of phenomenology: one that focuses on perception and

one on praxis. Each of these directions approaches the human-world relationship from a different side. Existential or 'praxis-oriented' phenomenology starts from 'the human side'. Its central question is how human beings act in their world and realize their existence. The main category here is *action*. Hermeneutic or perception-oriented phenomenology starts from 'the side of world', and directs itself at the ways reality can be interpreted and be present for people. The main category here is *perception*.

mediation of perception

The central hermeneutic question for a 'philosophy from the perspective of things' is how artifacts mediate human experiences and interpretations of reality. Don Ihde's philosophy of technology offers a good starting point for analyzing this, because of its focus on the technological mediation of perception. As Ihde shows, artifacts are able to mediate our sensory relationship with reality, and in doing so they transform what we perceive.

According to Ihde, the transformation of perception always has a structure of amplification and reduction. Mediating technologies amplify specific aspects of reality while reducing other aspects. When looking at a tree with an infrared camera, for instance, most aspects of the tree that are visible for the naked eye get lost, but at the same time a new aspect of the tree becomes visible: one can now see whether it is healthy or not. Ihde calls this transforming capacity of technology 'technological intentionality': technologies have 'intentions', they are no neutral instruments but play an active role in the relationship between humans and their world.

These intentionalities are no fixed properties of artifacts, however. They get shape within the relationship humans have with these artifacts. Within different relationships technologies can

have a different 'identity'. The telephone and the typewriter, for instance, were not developed as communication and writing technologies, and but as equipment for the blind and the hard of hearing to help them hear and write. In their use context they were interpreted quite differently, however. This phenomenon Ihde calls *'multistability:'* a technology can have several 'stabilities', depending on the way it is embedded in a use context. Technological intentionalities, therefore, are always dependent on the specific stabilities that come about.

Ihde's analysis of the transformation of perception has important hermeneutic implications. In fact, it shows that mediating artifacts co-determine how reality can be present for and interpreted by people. Technologies help to shape what counts as 'real'. This becomes most clear when investigating the role of instruments in the production of scientific knowledge. Without these, many scientific facts and theories could not exist. Instruments make it possible for scientists to perceive aspects of reality that cannot be perceived without them, like brain activity, microorganisms, or invisible forms of radiation emitted by stars. The 'reality' studied here, has to be 'translated' by technologies into perceivable phenomena. What 'reality' is in such situations is co-shaped by the instruments with which it is perceived.

Although this analysis of the technological mediation of perception and interpretation is not directly relevant within the context of ethics, the concepts used to understand this mediation are. Notions like mediation, amplification, reduction, intentionality and multistability can be translated to the context of the mediation of action, and therefore to the domain of ethics as well, as will become clear below.

mediation of action

Within the praxis-perspective, the central question is how artifacts mediate people's actions and the way they live their lives. While perception, from a phenomenological point of view, consists in the way the world is present for humans, praxis can be seen as the way humans are present in their world. The work of Bruno Latour offers many interesting concepts for analyzing how artifacts mediate action (Cf. Latour 1992; 1994). Latour points out that what humans do is in many cases co-shaped by the things they use. Actions are not only the result of individual intentions and the social structures in which human beings find themselves (the classical agency-structure dichotomy), but also of people's material environment. The concept introduced by Latour and Akrich to describe the influence of artifacts on human actions is 'script'. Like the script of a movie or a theater play, artifacts prescribe their users how to act when they use them. A speed bump, for instance, has the script 'slow down when you approach me;' a plastic coffee cup 'throw me away after use.'

This influence of artifacts on human actions is of a specific nature. When scripts are at work, things mediate action as material things, not as immaterial signs. A traffic sign makes people slow down because of what it signifies, not because of its material presence in the relation between humans and world. And we do not discard a plastic coffee cup because its user's manual tells us to do so, but because it simply is physically not able to survive being cleaned several times. The influence of technological artifacts on human actions can be of a non-lingual kind. Things are able to exert influence *as material things*, not only as *signs* or *carriers of meaning*.

As is the case with perception, in the mediation of action *transformations* occur. Following Latour, within the domain of action these transformations can be indicated as 'translations' of 'programs of action.' Latour attributes programs of actions to all entities -- human and nonhuman. When an entity enters a

relationship with another entity, the original programs of action of both are translated into a new one. When somebody's action program is to 'prepare meals quickly,' and this program is added to that of a microwave oven ('heating food quickly'), the action program of the resulting, 'composite' actor' might be 'regularly eating instant meals individually.'

In the translation of action, a similar structure can be discerned as in the transformation of perception. Just as in the mediation of perception some aspects of reality are amplified and others are reduced, in the mediation of action one could say that specific actions are 'invited,' while others are 'inhibited.' The scripts of artifacts suggest specific actions and discourage others.

The nature of this invitation-inhibition structure is as context-dependent as the amplification-reduction structure of perception. Ihde's concept of multistability also applies within the context of the mediation of action. The telephone has had a major influence on the separation of people's geographical and social context, by making it possible to maintain social relationships outside our immediate living environment. But it could only have this influence because it is used as a communication technology, not as the hearing aid it was originally supposed to be.

An important difference with respect to the mediation of perception, however, is the way in which action-mediating artifacts are present. Artifacts do not only mediate action from a ready-to-hand position (Ihde's 'embodiment relation') but also from a present-at-hand position (Ihde's 'alterity relation'). A gun, to mention an unpleasant example, mediates action from a ready-to-hand position, translating 'express my anger' or 'take revenge' into 'kill that person.' A speed bump, however, cannot be embodied. It will never be ready-to-hand; it exerts influence on people's actions from a present-at-hand position.

vocabulary

The STS concept of 'scripts,' indicating the influence of technological artifacts on human actions, can be seen as part of a more encompassing framework for understanding the role of technologies in the relation between humans and reality. The main concepts of this framework together form a 'vocabulary for technological mediation,' which could be helpful to analyze the role of technologies in their use contexts. Artifacts mediate perception by means of technological *intentionalities*: the active and intentional influence of technologies. They mediate action by means of *scripts*, which prescribe how to act when using the artifact. This latter form of mediation is most important for the ethics of engineering design, since it concern human *actions*, and ethics concerns the moral question 'how to act?'. Technological mediation appears to be context-dependent, and always entails a *translation* of action and a *transformation* of perception. The translation of action has a structure of *invitation* and *inhibition*; the transformation of perception a structure of *amplification* and *reduction*.

perception	**praxis**
mediation of perception	mediation of action
technological intentionality	script
transformation of perception	translation of action
amplification and reduction	invitation and inhibition
delegation: *deliberate inscription of scripts and intentionalities*	
multistability: *context-dependency of scripts and intentionalities*	

Figure 2: A vocabulary for technological mediation

3. Mediation and engineering ethics

The phenomenon of technological mediation creates a specific responsibility for designers. The brief overview of technical mediation in the previous section in fact shows that engineering design is an inherently moral activity. The fact that technologies-in-use inevitably play a constitutive role in the actions of their users places mediation in the heart of ethics. Ethics, after all, is about the question how to act -- and technologies appear to give 'material' answers to this question. Ethical questions regarding the design of technologies are thus not limited to questions about the goals for which technologies are designed and applied or to the quality of their functioning.

When technologies are inherently moral entities, this implies that designers are doing 'ethics with other means:' they

materialize morality. Usually, this 'doing ethics' happens in an implicit way. Engineers design a new technology with specific functionalities in mind, without explicitly aiming to influence the actions and behavior of users. The question, therefore, is how considerations regarding the mediating role that the technology-in-design will eventually play in society could be integrated in the design process.

There are two possible ways to take technological mediation into account during the design process. A first, minimal option is that designers could try to assess whether the product they are designing has undesirable mediating capacities. A second possibility goes much further: designers could also explicitly try to 'build in' specific desired forms of mediation.

3.1 Moralizing technologies

The latter direction was taken by the Dutch philosopher Hans Achterhuis (1995; 1998), who translated Latour's analysis of scripts into a plea for an explicit 'moralization of technology.' Instead of only moralizing other *people* ('do not shower too long;' 'buy a ticket before you enter the subway'), humans should also moralize their *material environment*. To a water-saving showerhead the task could be delegated to see to it that not too much water is used when showering, and to a tourniquet the task to make sure that only people with a ticket can enter the train.

Achterhuis' plea for a moralization of technology received severe criticism (Cf. Achterhuis 1998, 28-31). In the debate that arose around this issue in The Netherlands, two types of arguments were brought in against his ideas. Firstly, human freedom was thought to be attacked when human actions are explicitly and consciously steered with the help of technology. This reduction of human freedom was even perceived as a threat to human dignity; if human actions are not a result from deliberate decisions but from steering technologies, people were thought to

be deprived from what makes them human. $Moreover, if they are not acting in freedom, their actions cannot be called 'moral.' Human beings then simply show a type of behavior that was desired by the designers of the technology, instead of explicitly choosing to act this way. $Secondly, Achterhuis was accused to jettison the democratic principles of our society, because his plea for developing behavior-steering technology was considered an implicit propagation of technocracy. When moral issues are solved by the technological activities of designers instead of democratic activities of politicians, these critics hold, not humans but technology will be in control.

These arguments can be countered, though. Anticipating the mediating role of technologies during the design process -- either to assess possible undesired forms of mediation or to explicitly 'moralize' technologies -- does not need to be as immoral as it might seem to be. First of all, human dignity is not necessarily attacked when limitations of freedom occur. Our legal constitution implies a major limitation of freedom, after all, but this does not make it a threat to our dignity. Human behavior is determined in many ways, and human freedom is limited in many ways. Few people will protest against the legal prohibition of murder, so why protest to the material inhibition imposed by a speed bump to drive too fast at places where children are often playing on the pavement?

Secondly, the analysis of technological mediation above shows that technologies *always* help to shape human actions. Seen from this perspective, paying explicit attention to the mediating role of technologies should be seen as taking the responsibility that the analysis of technological mediation implies. When technologies are always influencing human actions, we had better try and give this influence a desirable form. Besides, as will become clear below in the example of a Dutch industrial design initiative, the 'moralizing' role of technologies does not necessarily have the form of exerting *force* on human beings to

act in specific ways. Technologies can also *seduce* people to do certain things; they can invite specific actions without forcefully exacting them.

These counterarguments, however, do not take away the anxiety that a technocracy would come about when technologies are explicitly moralized. It might be true that technologies do not differ from laws in limiting human freedom, but laws come about in a democratic way, and the moralization of technology does not. Yet, this does not justify the conclusion that it is better to refrain from paying explicit attention to technological mediation during the design process. If technologies are not moralized explicitly, after all, the responsibility for technological mediation is left to the designers only. Precisely this would amount to form of technocracy. A better conclusion would be that it is important to find a democratic way to 'moralize technology.' In the following, I will elaborate a way to do this.

3.2 Designing mediations

The moral impediments to the moralization of technology can be countered much easier than the practical impediments. The moralization of technological artifacts is not as easy as it might seem to be. In order to 'build' specific forms of mediation in technologies, designers need to anticipate the future mediating role of the technologies they are designing. And this is a complex task, since there is no direct relationship between the activities of designers and the mediating role of the technologies they are designing. As became clear above, the mediating role of technologies comes about in a complex interplay between technologies and their users.

Technologies are 'multistable,' as Don Ihde calls it. They have no fixed identity, but only get defined in their context of use. Technologies have to be 'interpreted' and appropriated by their users in order to be more than just objects lying around. Only

when human beings *use* them, artifacts become artifacts-for-doing-something. And this 'for doing something' is not entirely determined by the properties of the artifact itself, but also by the ways users deal with them, as became clear in the example of the telephone that was originally designed as a hearing aid. If this were not the case, accepting the idea of technological mediation would take us back to technological determinism; technologies would then be able to determine the behavior of their users all by themselves instead of being part of a sociotechnical network.

This multistability of technologies makes it very difficult to predict the ways in which technologies will influence human actions, and accordingly to evaluate this influence in ethical terms. Technologies can be used in unforeseen ways, and therefore have an unforeseen influence on human actions. Besides this, unforeseen forms of mediation can also emerge when technologies are indeed used as the designer intended. Revolving doors, for example, were designed to make it possible to enter a building while keeping the draught outside. Once they were used, they also appeared to inhibit people in a wheelchair to enter a building.

Designers thus help to shape the mediating roles of technologies, but these roles also depend on the ways in which the technologies are used and on the ways in which the technologies in question allow unforeseen mediations to emerge. The suggestion that 'scripts' are a result of 'inscriptions' (Akrich) or 'delegations' (Latour), therefore, does not do enough justice to the complex way in which mediation comes about. Designers cannot simply 'inscribe' a desired form of morality into an artifact. The mediating role of technologies is not only the result of the activities of the designers, who inscribe scripts or delegate responsibilities, but also depends on the users, who interpret and appropriate technologies, and on the technologies themselves,

which can evoke 'emergent' forms of mediation. The figure below draws together these factors in the coming about of mediation.

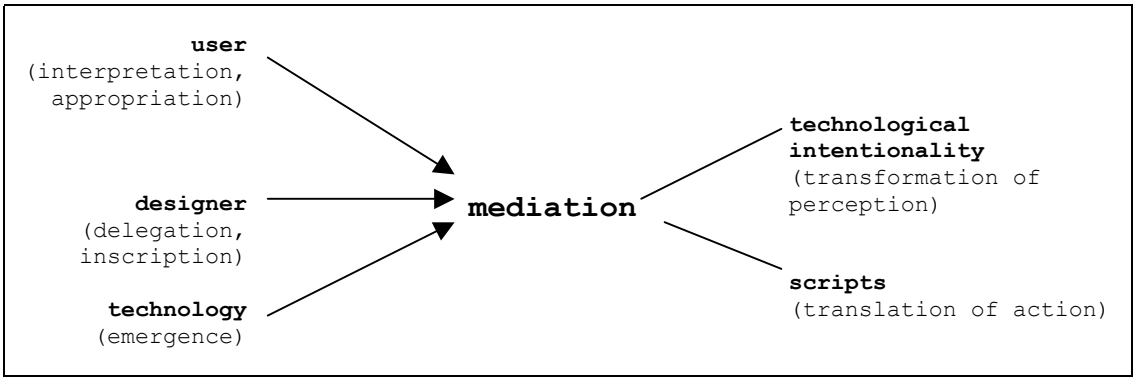


Figure 3: Sources of mediation

In order to cope with this complexity, designers should try to establish a connection between the context of design and the context of use. Designers could try to formulate product specifications not only on the basis of the desired functionality of the product, but also on the basis of an *informed prediction* of its future mediating role and a moral assessment of this role. Such a prediction could be indicated as a 'mediation analysis:' an analysis of the future role of the technology-in-design in terms of the vocabulary that was elaborated in section 2.

Mediation analyses could be carried out in two different ways. The first is simply: prediction by the imagination of the designer. Imagination can create a link between the designer and the technology in its future use context. When designers attempt to imagine what mediating role the technology they are designing might play in the behavior of its users, they could feed back these anticipations into the design process. As an example of this approach I will briefly discuss the work done by the Dutch industrial designers collective *Eternally Yours*. A second way to formulate an informed prediction of the future mediating role of

technologies is a more systematic one. It consists in an augmentation of the existing design methodology of Constructive Technology Assessment in such a way, that it becomes an instrument for a democratically organized moralization of technology.

3.3 Anticipation by imagination: 'Eternally Yours'

An interesting example of anticipating mediation by imagination is the work of the Dutch industrial designers collective *Eternally Yours*. Eternally Yours is engaged in eco-design, but in an unorthodox way (Cf. Van Hinte 1997; Verbeek 2000, 241-260). It does not want to address the issue of sustainability only in the usual terms of reducing pollution in production, consumption, and waste. The actual problem, Eternally Yours holds, is that most of our products are thrown away far before actually being worn out. Meeting this problem could be way more effective than reducing pollution in the different stages of products' life-cycles. For this reason, Eternally Yours focuses on developing ways to create product longevity. It does so by investigating how the coming about of attachment between products and their users could be stimulated and enhanced.

In order to stimulate longevity, Eternally Yours seeks to design things that invite people to use and cherish them as long as possible. '*It's time for a new generation of products, that can age slowly and in a dignified way, become our partners in life and support our memories*,' as Eternally Yours approvingly quoted the Italian designer Ezio Manzini in its letterhead. Eternally Yours investigates what characteristics of products are able to evoke a bond with their users. According to Eternally Yours, three dimensions can be discerned in the lifespan of products. Things have a technical, an economical and a psychological lifespan. Products can turn into waste because they simply are broken and cannot be repaired anymore; because they are outdated by newer models that have appeared in the market; and because

they do not fit people's preferences and taste anymore. For Eternally Yours, the psychological lifespan is the most important. The crucial question for sustainable design is therefore: how can the psychological lifetime of products be prolonged?

Eternally Yours developed many ideas to answer this question. For instance, it searched for forms and materials that could stimulate longevity. Materials were investigated that do not get unattractive when aging but have 'quality of wear.' Leather, for instance, is mostly found more beautiful when it has been used for some time, whereas a shiny polished chromium surface looks worn out with the first scratch. An interesting example of a design in this context is the upholstery of a couch that was designed by Sigrid Smits. In the velour that was used for it, a pattern was stitched that is initially invisible. When the couch has been used for a while, the pattern gradually becomes visible. Instead of aging in an unattractive way, this couch renews itself when getting old. Eternally Yours does not only pay attention to materials and product surfaces, however. It also investigated the ways in which services around products can influence their lifespan. The availability of repair- and upgrading services can prevent people from discarding products prematurely.

The most important way to stimulate longevity that should be mentioned in the context of this article, however, consists in designing products that evoke a bond with their users by engaging users in their functioning. Most technologies ask as little attention for themselves as possible when people are using them. Technologies, after all, are often designed to disburden people: a central heating system liberates us from the necessity to gather wood, chop it, fill the hearth, clean it, et cetera. We only need to switch a button and our house gets warm. But this disburdening character also creates a loss of 'engagement' with technological products. Ever fewer interactions are needed to use them (cf. Borgmann, 1992). One of the downsides of this

development is that this also affects the attachment between human beings and technological products. The product as a material entity has become less important than the function it fulfills. In many cases, human beings are not invited to interact with the technological artifact they are using, but only to consume the commodity it procures.

The work of Eternally Yours shows that this loss of engagement can be countered in a playful way. Technological products could invite users to interact with them without being so demanding that nobody would be prepared to use them. An interesting example in this direction is an engaging 'electric/ceramic heater' that was designed by Sven Adolph. It consists of a heating element with several concentric, cylindrically shaped ceramic shells of different height around it, that all have a vertical aperture. The shells can be arranged in several ways, so that they radiate their warmth in specific directions. This artifact is not a purely functional heater that withdraws into pure functionality like common radiators, that are hidden under the windowsill and are only turned on and off. It is an engaging product that asks for attention and involvement in its functioning, much like a campfire. You cannot hide it under the windowsill but have to put it in the middle of the room. You cannot escape it if you need warmth: you have to sit around it. Its shells have to be arranged if we want it to function. Simply turning the heater on and off is not enough: you actually have to be involved in its functioning if you want it to work.

The activities of Eternally Yours can be seen as a form of 'anticipating mediation by imagination.' Sigrid Smits' couch and Sven Adolph's heater were designed explicitly from the perspective of their possible mediating role in the interactions and affective relationships their owners will have with them. They mediate the behavior of their users in such a way that they are likely to get attached more to these artifacts than to other couches or heaters. These products were not only designed as

functional objects, but as artifacts that actively mediate the behavior of their users. The products of Eternally Yours embody an 'environmental ethics:' they seduce their users to cherish them rather than throwing them away prematurely.

3.4 Augmenting Constructive Technology Assessment

A second way to make an 'informed prediction' about the mediating role of a technology-in-design is a more systematic one. To establish a connection between the context of use and the context of design, designers could also employ a method that was developed precisely for making such a connection: the method of Constructive Technology Assessment (CTA) (cf. Schot 1992; Rip, Misa and Schot 1995). CTA creates a link between the contexts of design and use in a practical way: it aims to involve all relevant stakeholders in the design of technologies. In order to make use of the CTA methodology within the context of technological mediation, it needs to be augmented, though.

CTA is based on an evolutionary view of technology development. The process of technology development is seen as generating 'variations' that are exposed to a 'selection environment,' which is formed by entities like the market and government regulations. In this selection environment, only the 'fittest' variations will survive. There is an important difference between the generation of technologies and the generation of biological species, though. Contrary to biological evolution, in technology development there is a connection or 'nexus' between variation and selection. In order to prevent that much effort is put in developing technologies that will not be accepted by consumers or by government regulations, designers, after all, can anticipate the selection environment when they are designing technologies.

CTA is a method to employ this nexus in a systematical way, by feeding back assessments of the technology-on-design by all relevant actors -- like users, pressure groups, designers,

companies et cetera -- into the design process. It does so by organizing meetings of all relevant actors in which the aim is to reach consensus about the design of the technology that is 'constructively assessed.' This form of technology assessment is called 'constructive' because it does not assess technologies after they have been developed, but during their development, so that these assessments can be used to modify the original design. Besides this, CTA can be seen as a democratization of the designing process. When a CTA design methodology is followed, not only designers determine what a technology will look like, but all relevant social actors. Following this method, therefore, could take away the fear for technocracy that was discussed above.

Seen from the perspective of technological mediation, however, CTA also has limitations that need to be overcome. CTA primarily focuses on *human* actors, and pays too little attention to the actively mediating role of the *nonhuman* actor that is at the center of all activity: the technology-in-design. CTA claims to open the black box of technology by analyzing the complex dynamics of technology development. It bases itself on the constructivist notion that technologies are not 'given,' but the outcome of a process in which many actors are involved. Other interactions between the actors might have resulted in a different technology. But by analyzing the dynamics of *technology development* the black box of technology is only opened half way. It reveals how technologies emerge from their *design context*, but their role in their *use context* remains blackboxed. Therefore, organizing a democratically, domination-free discussion between all relevant actors is not enough to lay bare all relevant aspects of the technology in question. The mediating role of the technology-in-design is likely to remain hidden during the entire CTA process if it is not put explicitly and systematically on the agenda.

For this reason, participants in the CTA process should not only be invited to integrate assessments of users and pressure groups in product specifications, but also to anticipate possible mediating roles of the technology-in-design. The vocabulary for analyzing mediation, as presented in section 2 of this paper, could be a helpful for doing this. As the figure below illustrates, approaching the artifact-in-design in terms of mediation offers a perspective that can be used when creating a nexus between the contexts of design and use.

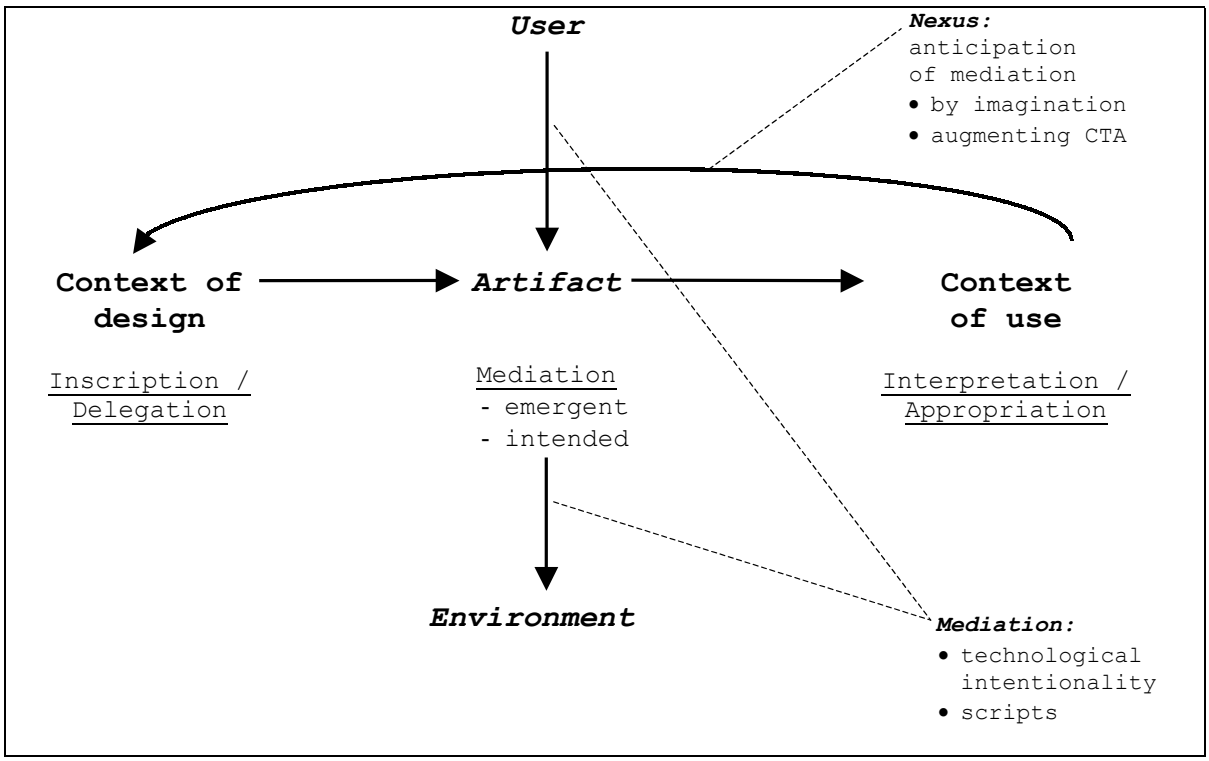


Figure 4: Mediation and the nexus between design and use

When the CTA method is augmented in this way, the method of 'anticipation by imagination' is given a more systematic character. Creating space for all relevant stakeholders to anticipate the possible mediating role of the technology-in-design enhances the chance that as many possible mediating roles

are taken into account. To be sure, this augmentation of the CTA methodology does not guarantee that all mediating roles of the technology in design will be predicted. It creates a connection between the 'inscriptions' within the context of design and the 'interpretations' or 'appropriations' within the context of use, but this cannot possibly cover all 'emergent' mediating roles of the technology. Yet, it might be a fruitful way to give shape to the responsibility of designers that becomes visible from the analysis of technical mediation.

4. Conclusion

The analyses of technological scripts and the mediating role of technologies, which have been elaborated over the past years in STS and the philosophy of technology, have major implications for the ethics of engineering design. The insight that technologies inevitably play a mediating role in the actions of their users makes the work of designers an inherently moral activity. Ethics is about the question how to act, and technologies appear to be able to give material answers to this question by inviting or even exacting specific forms of action when they are used. This implies that technological mediation should play an important role in the ethics of engineering design. Designers should not only focus on the functionality of technologies but also on their mediating roles. The fact that technologies always mediate human actions charges designers with the responsibility to anticipate these mediating roles.

This anticipation is a complex task, however, since the mediating role of technologies is not entirely predictable. In order to cope with this uncertainty, designers should try to bridge the gap between the context of use and the context of design. They can do so by using their imagination for carrying out a 'mediation analysis,' for which could this paper offers a vocabulary. They could also make used of an augmented form of

constructive technology assessment, in which the connection between design and use is not only made in imagination but also in practice. Designers then carry out a mediation analysis, together with all stakeholders that are involved with the technology-in-design, and decide in a democratically organized debate how to feed back the outcomes of this analysis into the design process. Following this method could also take away the fear that deliberately designing behavior-steering technology would lead to a technocratic society, since the inevitable mediating role of technology is then made subject to democratic decision-making. Analyzing technology in terms of mediation reveals that designing is materializing morality. This charges the ethics of engineering design with the task to profoundly rethink the moral responsibility of designers.

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NOTES

[1] The text of this section is a reworked version of a section in P.P. Verbeek, 'Acting Artifacts – the technological mediation of action,' forthcoming in P.P. Verbeek and A. Slob (eds.), Technology and Behavior (provisional title). It can be seen as a summary of the analysis technological mediation as developed in Verbeek, 2000.

[2] For the sake of clarity, I leave out a fourth human-technology relationship Don Ihde discerns: the 'hermeneutic relation'. For an analysis of this relation, see: Ihde, 1990.